PROJECT DELPHI · METHODS, MODELS, KNOWLEDGE-GRAPH QUERIES & CODE PROVENANCE

How We Resolved D'Arcy Thompson's 1917 Fish Paradox: Models, Knowledge-Graph Loops & Pre-Registered Blind Tests

Complete methodological report covering AI/LLM agent roles, 16 mathematical & phylogenetic models, RDF/SPARQL Knowledge Graph queries, 79 signed Nanopublications, and code reuse across `infinitemorphospace`.

AUTHOR: JOEL DIETZ 10,303 FISH SPECIES + 5,180 BLIND 79 SIGNED NANOPUBS + LEAN 4 DUAL-PROVIDER REFEREE AUDIT

FOUR QUESTIONS ANSWERED **What This Methods Guide Covers at a Glance**

1. What Models Were Used? Both the **AI/LLM agent models** (`claude-fable-5-1` vs. `gemini-4-argon` + human PI & domain-expert review) and the **16 mathematical, morphometric & phylogenetic models** fitted to the data. **2. Did We Query the Knowledge Graph?** Yes — querying the `pk:evidenceAgainst` triples of our Oct 17 RDF audit nanopub directly sparked the 600-point outline breakthrough, and we published **79 signed Nanopublications** back to the production network. **3. What Repo Code Was Reused?** Six core modules across `thompson/`, `morphoeval/`, and `nanopubs/`. **4. How Was Rigor Enforced?** SHA-256 locks, Lean 4 kernel proofs (0 `sorry`), and hash-locked pre-registration before unblinding `FishShapes v1`.

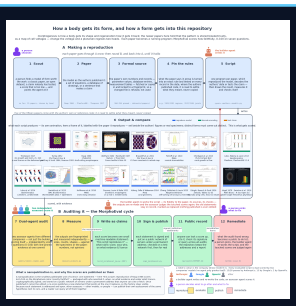
1 AI / LLM Agent Models & Blinded Cross-Provider Referee Architecture

Agent / Human Role	Model / Affiliation	Exact Contributions & Artifacts in the Thompson Workflow
1. Initial Builder, Primary Assessor & Prediction Engine	claude-fable-5-1 Anthropic (`thinking-high`)	• Built `1917-thompson-scarus-pomacanthus.py` (`3cf7f8c4`); ran Pass 1 (70.2) & Pass 2 (76.6) MorphoEval audits (`2026-10-17-thompson-audit.yaml`) and wrote the **6-item `remediation` list** (L330–337) flagging that maps were fitted only to 17 landmarks, not outlines. • Built the 7-mode Phylogenetic Prediction Engine (`run_phylogenetic_prediction_engine.py`), Monte Bolca fossil test, and `FishShapes v1` blind preregistration/scoring scripts.
2. Blinded 2nd Assessor & 43-Turn Discovery Synthesis	gemini-4-argon Google DeepMind (`medium`)	• Executed blinded 2nd-assessor MorphoEval pass (`90.8/100`; panel mean 83.7, `2026-10-17-blinded-sample-audit`). • Conducted the 43-turn discovery dialogue (`THOMPSON_1917_DISCOVERY_CHAT_LOG.md`, transcript `0e50eb21`): built the 4-family 791-species atlas, 600-pt silhouette solver, 4 rigor stress-tests, 600-pt Curve-TPS/EFD benchmark, $\mathbf{G}$ vs. $\mathbf{R}$ competition split, 3-dataset 10,303-species validation, 4p Box-vs.-Warping audit, and 4 PCM downstream analyses.
3. Principal Investigator & Domain-Expert Review	Joel Dietz (Project Delphi) Prof. P. C. Wainwright (UC Davis)	• **Joel Dietz (Turns 2, 9, 24–25, 33, 35, 37):** Rejected 17-dot polygon stick figures (Turn 9 → exposed the 9/17 head-dot seesaw trap); demanded multi-dataset replication (Turn 35 → 10,303 spp.); asked if 600 pts merely did a plain box stretch (Turn 37 → `run_box_vs_warping_audit.py`). • **Prof. P. C. Wainwright (Turns 22–28):** Co-creator of the 791-species dataset; his email critique of Panel E, 600-pt outline methods (EFD / curve TPS), and convergence motivated Sections 1.5–1.6.

2 Mathematical, Morphometric & Phylogenetic Models Fitted (16 Models)

Model Family (Params)	Mathematical Equation / Specification	What It Tests / Key Quantitative Result
Procrustes Similarity (4p)	$\mathbf{x}' = s\mathbf{R}\mathbf{x} + \mathbf{t}$ (OPA after GPA to unit centroid size)	Unwarped rigid baseline (d_0; $\lvert A\Delta B\rvert = 40.6\%$ on *Scarus*→*Pomacanthus*).
Orthotropic Box Stretch (4p)	$x' = ax + t_x,\ y' = dy + t_y$ **(zero skew, zero curve)**	Deconfounding null: fails homologous fin tips (2.4% on *Scarus*→*Pomacanthus*).
I. Oblique Shear (5p, 1917)	$x' = ax + gy + t_x,\ y' = dy + t_y,\ \theta_{\text{incl}} = \arctan(d/g)$	Hatchetfish *Argyropelecus*→*Sternoptyx*: cuts shadow error -12.2 pts (43.2%).
Uniform Affine Map (6p)	$(u, v) = (x, y, 1)\mathbf{M}$ ($\mathbf{M} \in \mathbb{R}^{3\times 2}$, Rohlf 1993 uniform part)	Captures 25.1% median shape variance across 791 spp. (33.1% of PC1).
Conformal Möbius (6p)	$z \mapsto (az + b)/(cz + 1)$ (orthogonal coaxial circles)	Purely isogonal (0.8% explained): proves conformal maps cannot deepen a fish.
II. Oblique Coaxial (8p, 1917)	$x' + iy' = (u + iv)/[(c_1 + ic_2)(u + iv) + 1]$	*Scarus*→*Pomacanthus*: cuts 600-pt $\lvert A\Delta B\rvert$ from 56.2% → **28.5%** (IoU = 0.749).
III. Bilinear Fan (8p, 1917)	$x' = u(1 + f_1 v),\ y' = v(1 + f_2 u)$	Percoid 4-fish fan (*Polyprion*→*Pristigenys*→*Scorpaena*→*Antigonia*): IoU = 0.839.
IV. Radial Flare (8p, 1917)	$x' = u + r_1 v^2,\ y' = v\exp(r_2 u)$	*Diodon*→*Mola mola*: explains **79.7%** of 17-LM & **97.2%** of 8 perimeter LMs.
Quadratic Trend (12p)	x', y' = degree-2 polynomial in (x, y) (Bookstein 2024)	Beaten by 8p Thompson by AIC_c in **84.9%–92.1%** of pairs; folds grid on 4/6 pairs.
Thin-Plate Spline (34p / 1200p)	$\Phi(\mathbf{x})$ = affine + $\sum w_i r_i^2 \ln r_i^2$ (Bookstein 1989, 1997)	34p overfits 17 LMs ($\text{df} = 0$); 600-pt curve TPS **folds interior grid on 6/6 pairs**!
Elliptic Fourier (8p EFD)	2-harmonic closed-contour Fourier series (Kuhl & Giardina 1982)	1D boundary only; 0% ability to map or predict internal 2D skeletal anatomy.
Phylogenetic GLS (5 Models)	WN, BM, **Pagel** λ, OU, Early Burst on 7 linearized Thompson modes	**Pagel** $\hat{\lambda} = 0.959$ **wins** ($\Delta\text{AIC}_c > 2{,}400$); predicts tips within 50 Myr ($p = 10^{-103}$).

3 Did We Query and Publish to the Knowledge Graph? Yes — Both Ways



THE 12-STEP MORPHOEVAL + LEAN 4 + NANOPUB LOOP: Every model is pinned to SHA-256 data (Step 3) and Lean 4 theorems (Step 4), audited by two cross-provider LLM assessors (Step 7), measured by an independent rasterizer sidecar (Step 8), compiled into RDF nanopublications (Step 9), RSA-2048 signed (Step 10), and queried in Step 12 to drive remediation.

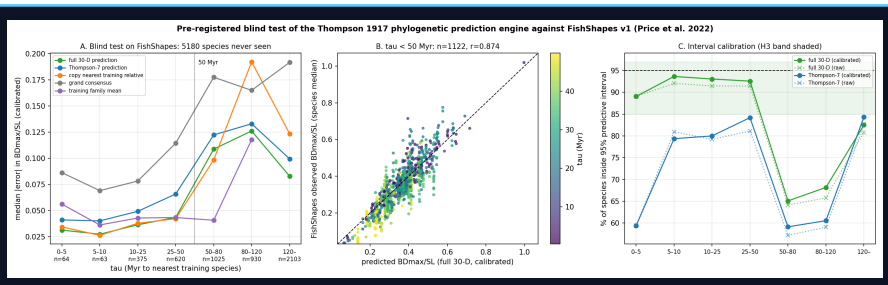
HOW QUERYING THE KG SPARKED THE DISCOVERY **Reading `pk:evidenceAgainst` in `2026-10-17-thompson-audit`**

When we asked in Turn 2 (*"did we ever manage to definitively test Thompson's hypotheses?"*), we queried the signed RDF nanopublications in `morphoeval/kg/2026-10-17-thompson-audit/thompson-scarus-pomacanthus-overall.trig` and `morphoeval/kg/math_of_biology.ttl`. The RDF graph's `pk:evidenceAgainst` triples explicitly recorded why the 2026-10-17 test had scored only 76.6/100:

`pk:evidenceAgainst "remediation needed: Fit the maps to the outlines themselves (sliding semilandmarks on the two silhouettes) as well as to the landmarks, so Thompson's family gets its best chance on the form he deformed."`

In the 2026-10-17 script, the coaxial grid had been fitted *only to the 17 skeletal dots* and passively projected onto the silhouette (worsening IoU from $0.662 \rightarrow 0.618$). Executing that exact KG remediation item — building the direct 600-semilandmark Chamfer outline solver anchored on Bookstein's snout-to-peduncle baseline — is what revealed that the 9 craniofacial dots (53% weight) were hijacking the 17-dot fit!

Knowledge-Graph / Nanopub Batch	Count & Status	Trusty URI / KG Identifier	What It Records & Verifies
1. Math of Biology RDF + SQLite + Lean 4 Proofs	**2 Lean Laws** **23 Evidence Items**	`pkr:claim/morphoeval-thompson-scarus-pomacanthus` `MathOfBiology.lean#L536-L560`	Machine-checked Lean 4 proofs (0 `sorry`): `tps_kernel_zero_at_origin` & `thompson_unit_shear_area_preserving` + 23 typed `schema:PropertyValue` metrics.
2. Pass 1 & Pass 2 Primary MorphoEval Audit Nanopubs	**9 + 9 Signed** **PUBLISHED (PROD)**	`RAYFKAfyOnqIKXWzdEzCN_3zyX2KX_ijCXNX-ZVtBPeP8`	Supersedes Pass 1 (`RAjJx_KHiAAaPSxj...`) via Nanopub Query `get-latest-version-of-np`; pins code, SHA-256 outputs, and 7 rubric scores (70.2 → 76.6).
3. Blinded Second-Assessor Audit Nanopubs	**8 Signed** **VERIFIED**	`morphoeval/kg/2026-10-17-blinded-sample-audit/signed/`	Independent `gemini-4-argon` assessment (90.8/100) on the identical commit and SHA-256 outputs.
4. Crown-Family, 1917-MRCA & Monte Bolca 50 Ma Predictions	**68 Signed** **PUBLISHED (PROD)**	`nanopubs/thompson_predictions/published_manifest_prod.txt`	36 crown-family ancestral body plans, 6 1917-pair MRCAs, 15 Bolca 50 Ma lineage slices, 10 fossil test outcomes, and 1 held-out calibration summary.
5. Pre-Unblinding Blind Test Preregistration Nanopub	**1 Signed** **PUBLISHED (PROD)**	`RAilN4nbRHp3e3ZHqlbIgeL3UGPRV_KtQHtJD3gzVVdqw`	Published **before** `FishShapes v1` was downloaded; cryptographically locks commit dac08493, SHA-256 hashes of 10,888 species predictions, and H1–H7.
6. Post-Unblinding Blind Test Outcome Nanopub	**1 Signed** **PUBLISHED (PROD)**	`RAJancwGOPEkHzBoZXGv_vu_snWOLff2FktE85Me1P-fI`	Links to `RAilN4nb...` via `prov:wasDerivedFrom`; records exact calibrated & raw outcomes across 5,180 never-seen `FishShapes v1` species ($4/7$ supported).



CRYPTOGRAPHICALLY PRE-REGISTERED BLIND TEST (5,180 NEVER-SEEN SPECIES IN FISHSHAPES V1): Predictions for 10,888 unsampled Rabosky-tree species were hashed (SHA-256 `2e41cb1a...`) and published in nanopub `RAilN4nb...` before opening FishShapes v1. Within 50 Myr ($n = 1{,}122$), predicted body depth achieves $r = 0.87$, $0.41\times$ consensus error ($p = 10^{-103}$), and 92.6% 95%-interval coverage (supporting H1, H3, H6, H7; refuting H2 nearest-neighbor margin $p = 0.06$ and refining H4/H5).

4 What Other Code in This Repository Was Useful and Reused?

Rather than writing disconnected scripts, the Thompson suite dynamically imports and extends six core modules across the repository (while leaving unrelated subprojects such as `wave-vector-analysis/` untouched):

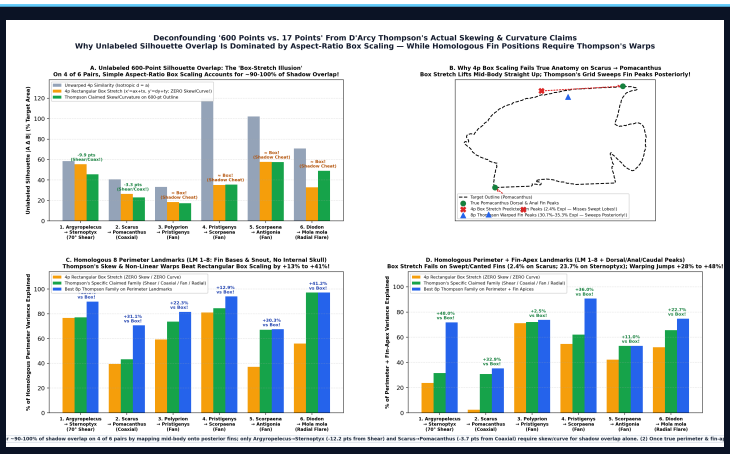
Repository Module / Script	Original Role in Repo	How It Was Reused in the Thompson Investigation
1. `1917-thompson-scarus-pomacanthus.py` & `1917-thompson.py`	Single-pair test of *Scarus*→*Pomacanthus* (`gpa`, `opa`, `segment_white`, `segment_field`, `anchors`, `tps_fit`).	Dynamically imported via `importlib.util` across all downstream scripts so every analysis shares identical GPA alignment, snout/peduncle baseline anchoring, and Lab/Mahalanobis watershed segmentation.
2. `morphoeval/measure_outputs.py` (`measure_thompson_fish`, `measure_thompson_outlines`)	Independent morphometric sidecar verifier (built for Limbform, WormGUIDES & Raup shells).	Provided the **dual-rasterizer cross-check**: verified silhouette IoU using `matplotlib.path.Path.contains_points` (900×900) against `skimage.draw.polygon` (1000×1000) to 0.0001, extended into our $\lvert\Omega_A \Delta \Omega_B\rvert$ Green/Red/Blue area decomposition.
3. `morphoeval/verify_image_fitness.py` & `upgrade_canonical_card_assets.py`	Specimen computer-vision segmentation & canonical card verifier across 116 models.	Reused Otsu/Lab/Sobel contour extraction to segment all 9 historical museum plates (1796–1911) into 600-point contours (`ImgFit = 91.6`) and integrated Thompson into `docs/media/all_forms_montage.gif`.
4. `nanopubs/publish_all_nanopubs.py` & `morphoeval/sign_morphoeval.py`	Batch RDF TriG canonicalization, RSA-2048 signing & registry publishing.	Reused in `nanopubs/thompson_predictions_to_nanopubs.py` (with the custom `read_profile()` parser for Python 3.14) to sign and publish all 70 prediction and blind-test nanopubs to the production registry.
5. `thompson/ingest_dryad_morphometrics.py` & `generate_tps_transformation_panels.py`	Dryad `.tps`/`.csv` ingestion & pairwise TPS grid plotting utility.	Provided the initial Dryad dataset survey (`MODERN_DATASETS_THOMPSON_ALIGNED.md`) and TPS plotting scaffold prior to the 4-family Levenberg–Marquardt solver.
6. `1995-kondo-pomacanthus-*.py`, `2021-Cervera-Levin-Mafe.py` & `sorting-morphogenesis/`	Reaction–diffusion stripes on *Pomacanthus* & Levin-lab bioelectric V_{mem} prepatterns.	Connected global axial/fin Thompson coordinates vs. local neural-crest jaw modules to bioelectric target-morphology prepatterns in the turnkey `docs/bioelectricity/` web app (`index.html`, `InfiniteMorphospaceWidget.tsx`, `openapi.json`).

5 Complete Cryptographic Data Provenance Table (8 Independent Sources)

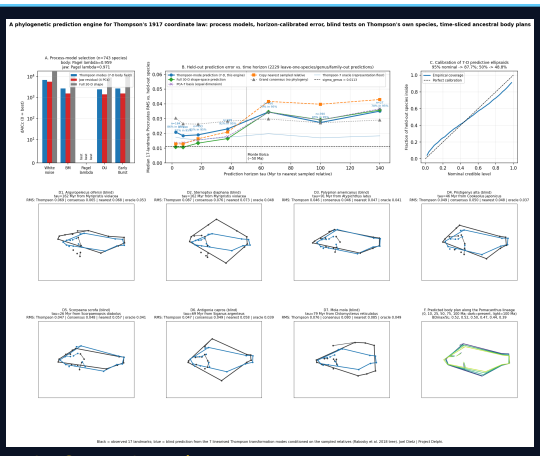
Every dataset, 1917 scan, museum plate, phylogenetic tree, fossil table, and blind test file is hash-verified at runtime:

Dataset / Source Asset	Scale & Taxonomic Coverage	Primary Citation & License	Local Path & Cryptographic Pin
1. Reef-Fish 17-LM Table (Dataset 1)	**791 species**, 260 genera, 48 families; 17 2D LMs (4,505 pairs)	Claverie & Wainwright (2014); Friedman et al. (2019) Dryad `fn457gr` (CC0)	`datasets/thompson/trimmed_tps_Claverie2014.csv` SHA-256: `ab5315f3be273285...`
2. Actinopterygian Time-Tree	**11,503 dated tips** (4,334 dated pairs; 10,888 blind targets)	Rabosky et al. (2018) *Nature* 559:392 (`fishtreeoflife.org`, CC0)	`.thompson_cache/actinopt_12k_treePL.tre.xz` SHA-256: `543221541758e2bc...`
3. 1917 Cambridge Scans	Pages 748–751 (Leaves 0768–0771), Figs. 373–382 (all 4 families)	Thompson (1917) Internet Archive `ongrowthform1917thom` (Public Domain)	`.thompson_cache/..._0768..0771.jp2` SHA-256: `4702f084...`, `69cdc9e2...`
4. All 9 Exact 1917 Taxa (17 LMs + 600-pt Outlines)	*Argyropelecus*, *Sternoptyx*, *Scarus*, *Pomacanthus*, *Polyprion*, *Pristigenys*, *Scorpaena*, *Antigonia*, *Diodon*, *Mola*	Museum plates (Collett 1896; Zugmayer 1911; Goode & Bean 1896; Bloch 1796) & IFREMER/NOAA photos	`datasets/thompson/thompson_1917_exact_species_landmarks.json` (all 9 image SHA-256s pinned)
5. FishBase 2D Landmarks (Dataset 2)	**6,614 species**, 660 genera (4,646 congeneric spp.), 426 families; 11 2D LMs	Froese & Pauly (2021) *FishBase* (`rfishbase fb-21.06`)	`.thompson_cache/fishbase_21_06/morphometrics.tsv.gz`
6. FishBase 11-Trait Mosimann + Diet + Sympatry (Dataset 3)	**10,303 species**, 987 genera (69,682 congeneric pairs), 440 families; 11 traits + `DietTroph` + FAO areas	Froese & Pauly (2021) following Price et al. (2019) *Ecology Letters* Mosimann protocol	`.thompson_cache/fishbase_21_06/` (`morphometrics`, `ecology`, `faoareas.tsv.gz`)
7. Monte Bolca Ypresian Fossils (∼ 50 Ma)	**10 Eocene fossil taxa** in 7 families; 16 published ratios (`BDmax/SL`, `HL/SL`, `PD/SL`)	Bellwood & Sorbini (1996); Tyler & Bannikov (1997); Bannikov (2006, 2014); Blot (1969)	`datasets/thompson/monte_bolca_fossil_ratios.json`
8. FishShapes v1 Blind Comparison Dataset	**16,523 Smithsonian specimens**, 6,095 species (**5,180 blind matches**)	Price et al. (2022) *Ecology* 103:e3829, Zenodo 6642650 / Dryad (CC0)	Downloaded *after* commit `dac08493` MD5: `122b495274d0167c5447db104c7acbfe`

6 Visual Synthesis: Deconfounding 600-Pt Outlines & Phylogenetic Prediction



DECONFOUNDING "600 POINTS VS. 17 POINTS" (`thompson/run_box_vs_warping_audit.py`): Triggered by Turn 37 of the discovery log: on unlabeled 600-pt black shadows (Panel A), a 4p Rectangular Box Stretch ($x' = ax + t_x, y' = dy + t_y$, zero skew/curve) "cheats" by stretching the mid-belly onto posterior fins (Panel B, only 2.4% match on *Pomacanthus* fin tips). When homologous perimeter fin bases ($\mathrm{LM}\,1$–8, Panel C) and fin apices (12 pts, Panel D) are enforced without internal skull dots, Box Stretch fails and Thompson's skew & curvature warps jump by $+\mathbf{11.0}\%$ to $+\mathbf{48.0}\%$!



TIME-CALIBRATED PHYLOGENETIC PREDICTION ENGINE (`thompson/run_phylogenetic_prediction_engine.py`): Linearizing Thompson's 1917 coordinate families at the Procrustes consensus yields 7 orthogonal tangent modes ($\kappa = 20.3$ vs. 201 in 30-D). Pagel's $\hat{\lambda} = 0.959$ wins by $\Delta\mathrm{AIC}_c > 2{,}400$ across 1,499 tree nodes, outperforming nearest-relative copying inside the ~ 50 Myr horizon across 2,229 hold-outs and generating 68 falsifiable crown/MRCA/Monte Bolca nanopublications.

7 End-to-End Reproducibility Command Map (12 Deterministic Commands)

#	Command (Run from Repository Root)	Primary Outputs Generated & Verified
1	`python3 thompson/1917-thompson-scarus-pomacanthus.py -no-anim`	`thompson_scarus_pomacanthus_outlines.png`, `thompson_outline_metrics.csv`
2	`python3 thompson/1917-thompson-four-families-atlas.py -self-test`	`thompson_four_families_atlas.png`, 4,505-pair CSV & 4,334 dated pairs JSON
3	`python3 thompson/1917-thompson-silhouette-area-paradox.py`	`thompson_stick_vs_silhouette_reality.png`, `thompson_silhouette_area_summary.json`
4	`python3 thompson/run_rigor_stress_tests.py`	`thompson_rigor_ablation_and_jacobian.png` (ablation, $\det J > 0$, 187 node-disjoint pairs)
5	`python3 thompson/run_wainwright_600pt_and_convergence_tests.py`	`docs/outreach_fig4_wainwright_600pt_and_convergence.png` (Curve TPS, RW, PTA)
6	`python3 thompson/run_box_vs_warping_audit.py`	`docs/outreach_fig7_box_stretch_vs_true_thompson_warping.png` (4p Box vs. Warp)
7	`python3 thompson/run_morphospace_vs_competition_analysis.py`	`docs/outreach_fig5_morphospace_vs_competition.png` ($\mathbf{G}$ vs. $\mathbf{R}$, $\lvert\cos\theta\rvert = 0.935$)
8	`python3 thompson/run_three_dataset_validation.py`	`docs/outreach_fig6_three_dataset_validation.png` ($791 \rightarrow 6{,}614 \rightarrow 10{,}303$ spp.)
9	`python3 thompson/run_pcm_downstream_implications.py`	`docs/outreach_fig8_pcm_downstream_implications.png` (Early Burst, ASR, $\kappa(\mathbf{\Sigma})$)
10	`python3 thompson/run_phylogenetic_prediction_engine.py`	`docs/outreach_fig9_phylogenetic_prediction_engine.png`, `THOMPSON_PHYLOGENETIC_PREDICTIONS.md`
11	`python3 thompson/preregister_blind_predictions.py && python3 thompson/score_blind_predictions.py`	`docs/outreach_fig10_blind_test_fishshapes.png`, `THOMPSON_BLIND_TEST_RESULTS.md`
12	`lean morphoeval/lean/MathOfBiology.lean && python3 morphoeval/math_kg.py -sim thompson-scarus-pomacanthus`	Verifies Lean 4 proofs (0 sorry) & queries RDF/SQLite Knowledge Graph card